# Thank you!

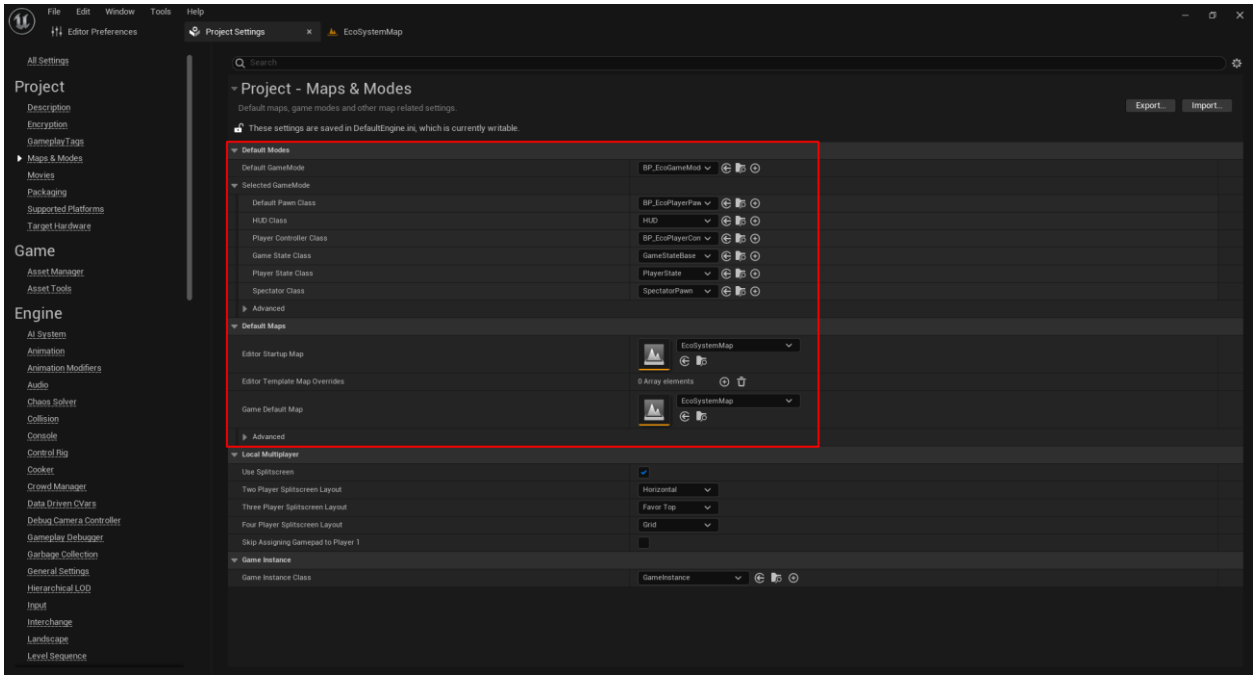
Thank you for purchasing the Ecosystem Simulation Starter Pack. This pack has several features that you can change at will without interfering with other functions. I tried to make it as modular and versatile as possible.

This pack is still under development. Based on the feedback from my beta testers and users like you, I will try to constantly update it to the maximum functionality. Future updates will be available to anyone who has purchased this asset on the Epic Marketplace.

If you have any questions, concerns, or need assistance, please feel free to email me at stylus219@gmail.com.

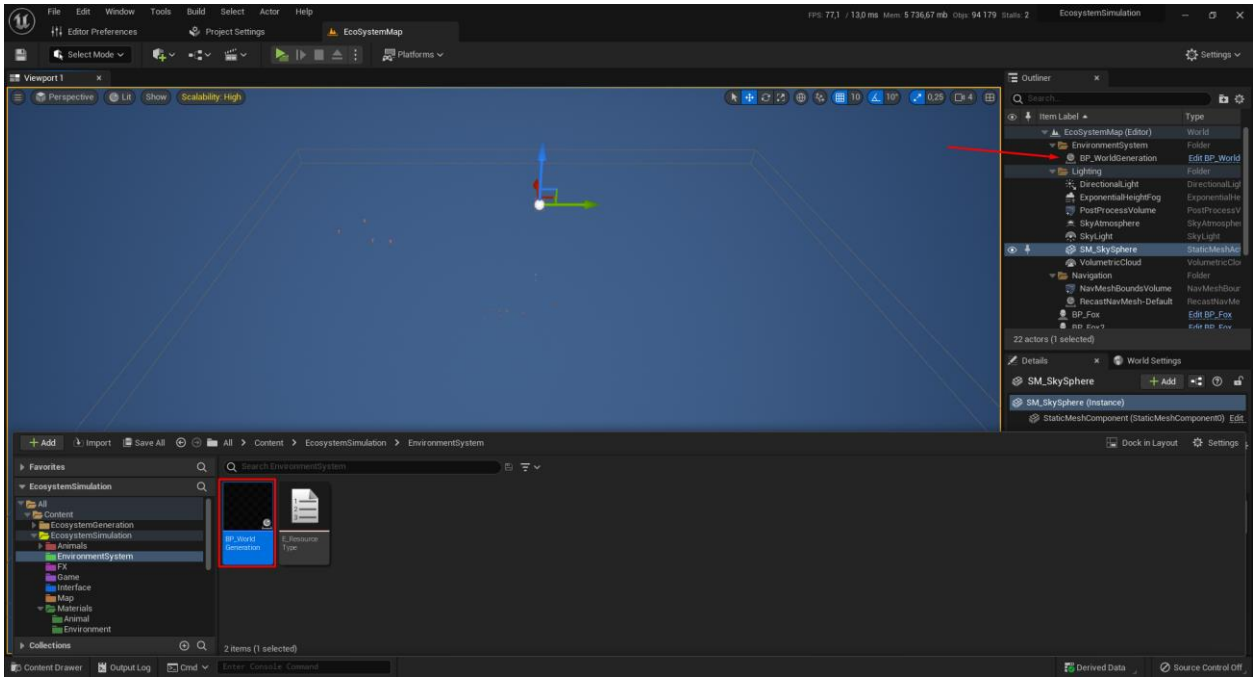
## PART 1. ECOSYSTEM WORLD GENERATOR

1.1. First of all, to use our assert, you need to enable GameMode

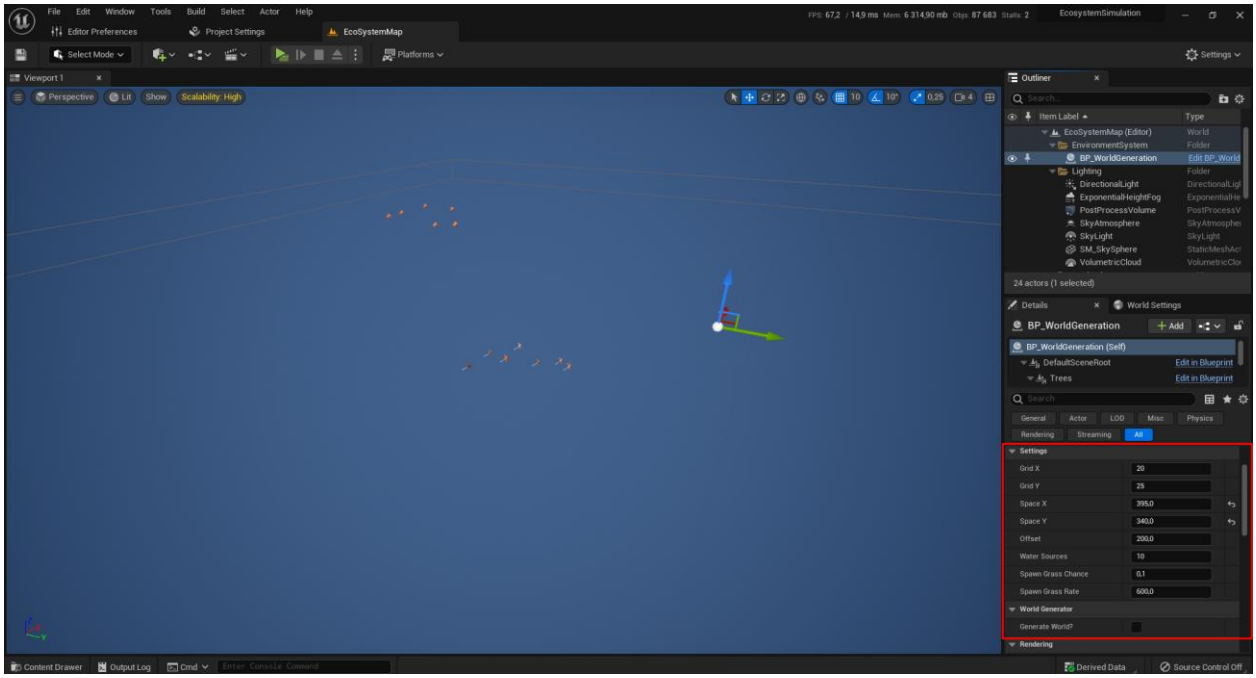



This is a standard GameMode for this project, you can use your own.

1.2. Now you need to add **BP_WorldGeneration** to the map if it is not there.




1.3. After adding **BP_WorldGeneration** to the map, we will have access to the generator settings.



Main parameters:

**GridX** - *number of tiles on the X-axis*

**GridY** - *number of tiles on the Y-axis*

Since we are using a hexagonal grid, we need to have a few auxiliary parameters, if you want to use your own tiles or resize existing ones, you will need to adjust the offset parameters:

**SpaceX** - *offset of the tile along the X-axis*

**SpaceY** - *offset of the tile along the Y axis*

**Offset** - *the offset of the tile along the X axis, after the offset along the Y axis and every second step*

**WaterSources** - *number of tiles with fresh water*

**SpawnGrassChance** - *chance of sleeping grass (food for rabbits) on each green tile (from 0 = 0% to 1 = 100%)*

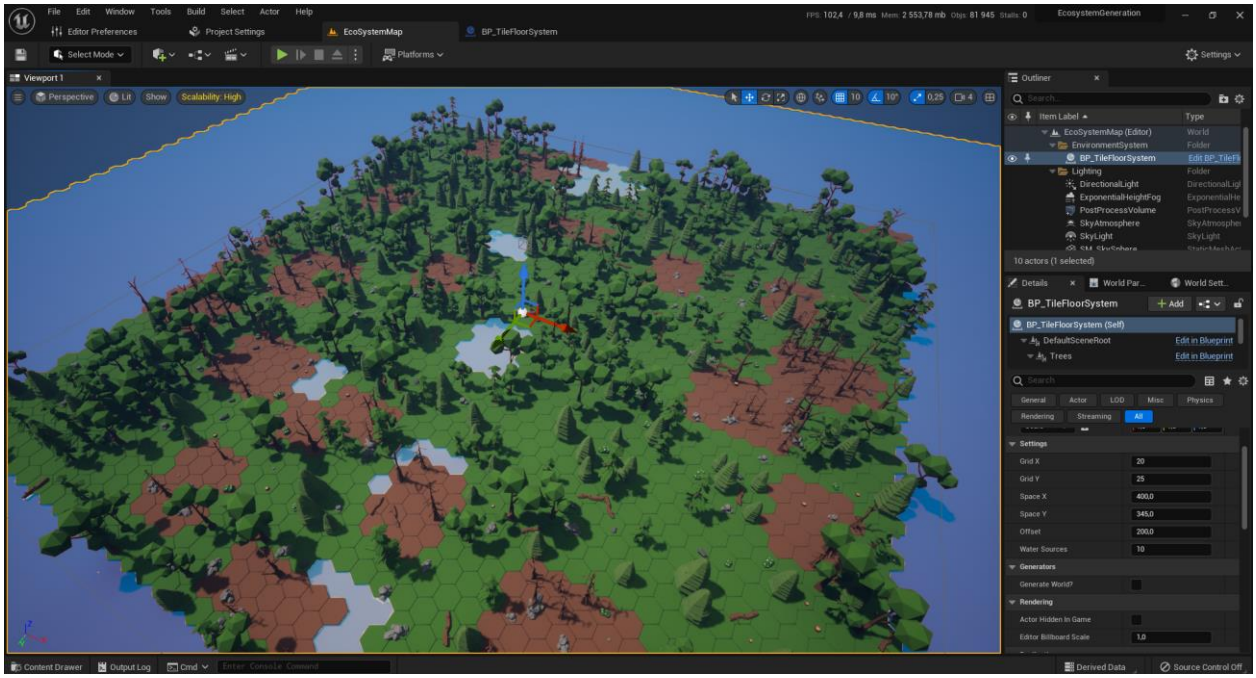
**SpawnGrassRate** - *The time after which grass with the "SpawnGrassChance" chance is added to the map.*

After setting up, you need to check the box;

**GenerateWorld?**

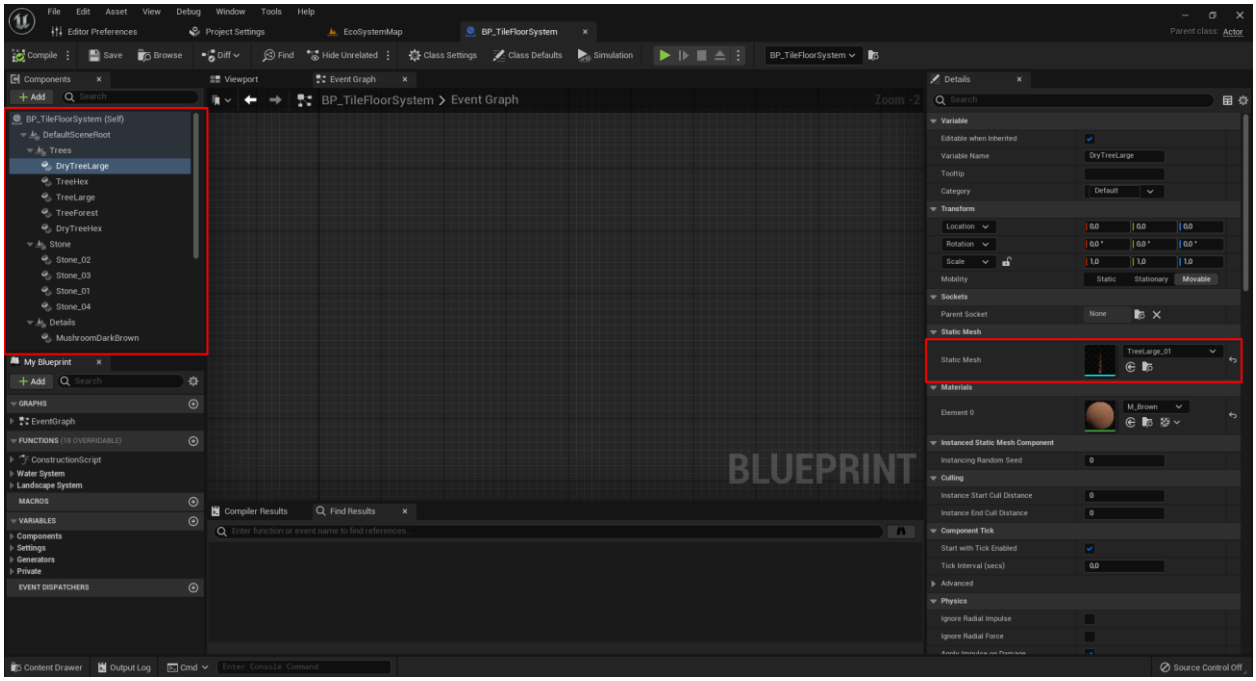
(It will automatically be set to **false** after generation, which will allow you to generate the world again)

1.4. After clicking **GenerateWorld?** we see the generated world, each time it will be different.



You can also use your own land tiles. Other models of trees and various details on the map.

To do this, just go to the **BP_WorldGeneration** blueprint:



Here we see on the left the instances we use for generation.

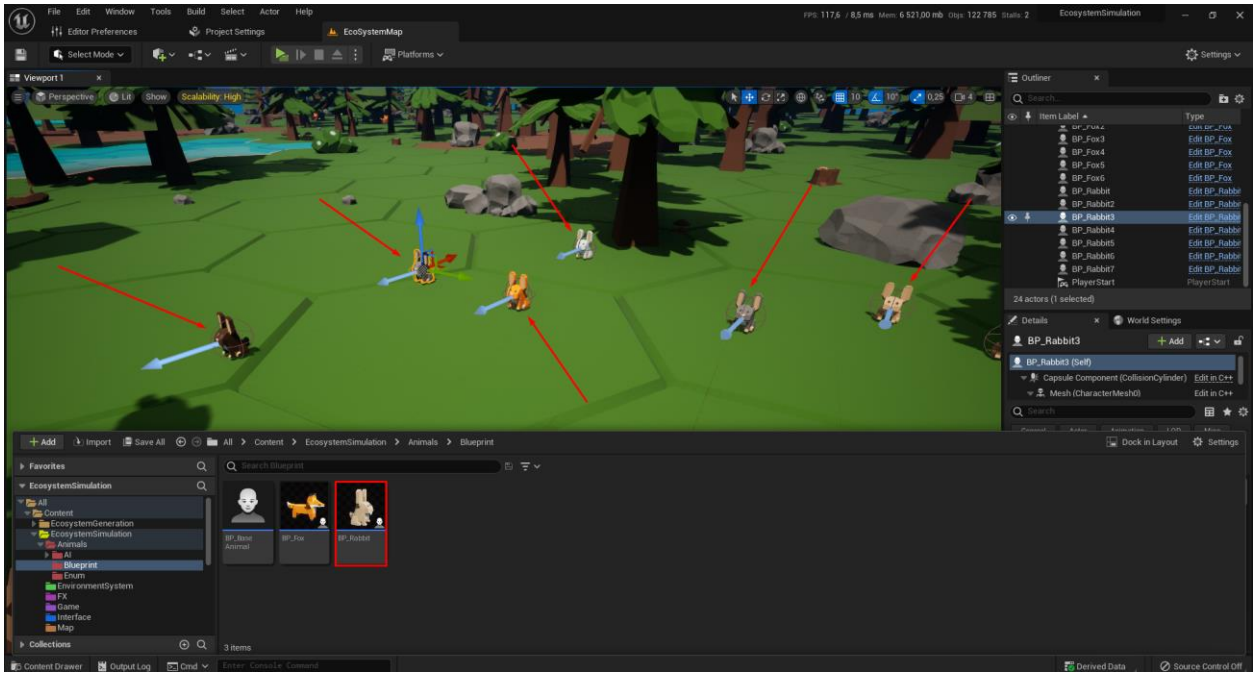
And on the right, we can replace the static models for these instances.

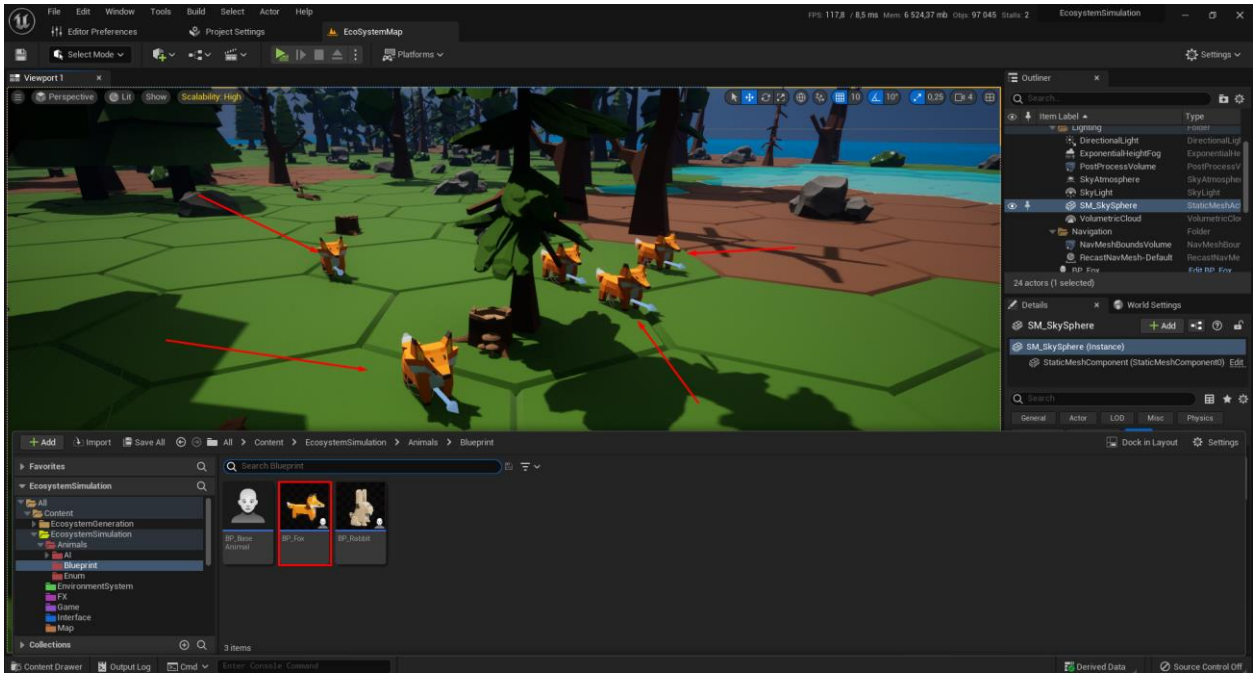
## PART 2. ECOSYSTEM ANIMAL AI

### 2.1. Herbivores and carnivores

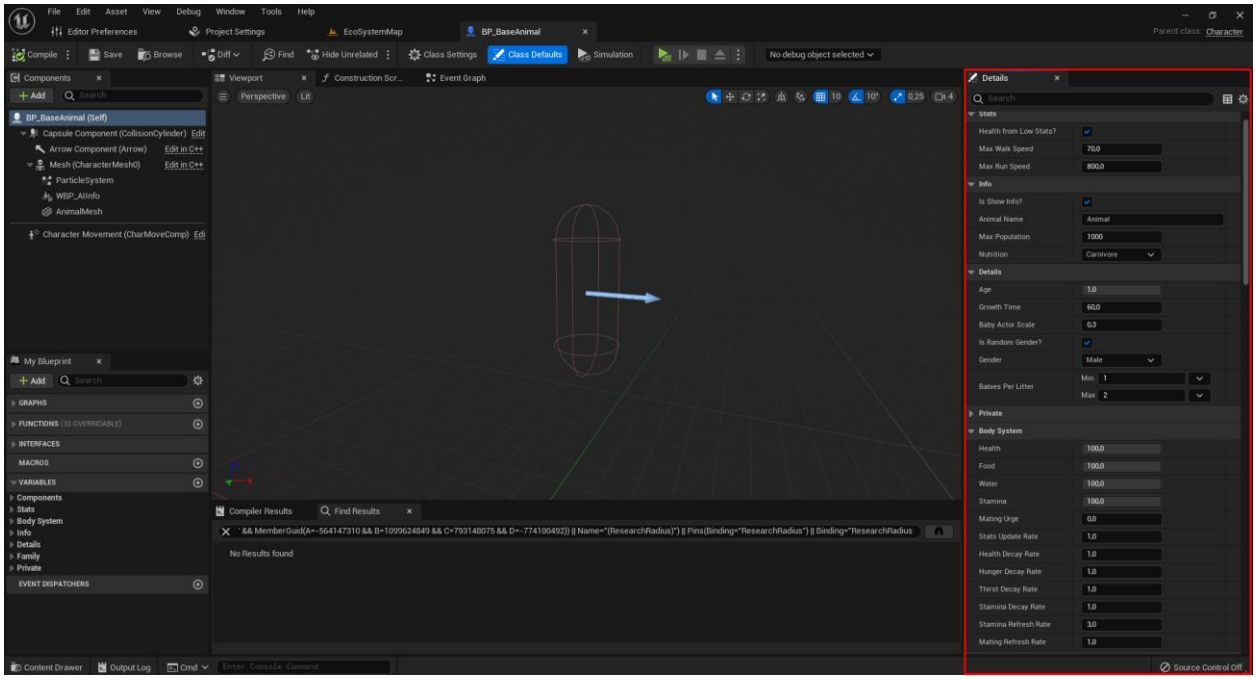
At this point, we have herbivorous rabbits and carnivorous foxes to add to the map.

In this case, we create a food chain (fox -> rabbit -> grass)





2.2. Setting the base class of the animal.



In this step, I'll talk about the parameters of the base animal class, which will help you understand what parameters to adjust for child classes.

**STATS:**

**HealthFromLowStats?** - *a parameter that enables health reduction if the animal is hungry or thirsty;*

**MaxWalkSpeed** - *Maximum speed of calm movement;*

**MaxRunSpeed** - *Maximum running speed;*

**BODY SYSTEM:**

**Health** - *standard/maximum health value (if 0 animal dies);*

**Food** - *standard/maximum value of stomach filling with food (if 0 start "HealthDecayRate");*

**Water** - *standard/maximum value of stomach filling with water (if 0 start "HealthDecayRate");*

**Stamina** - *standard/maximum stamina value (if 0 start sleeping "StaminaRefreshRate");*

**MatingUrge** - *attraction to the opposite sex (at the start 0);*

**StatsUpdateRate** - *frequency of updating animal statuses (an important parameter, the variables below will be executed with the frequency you set in this parameter);*

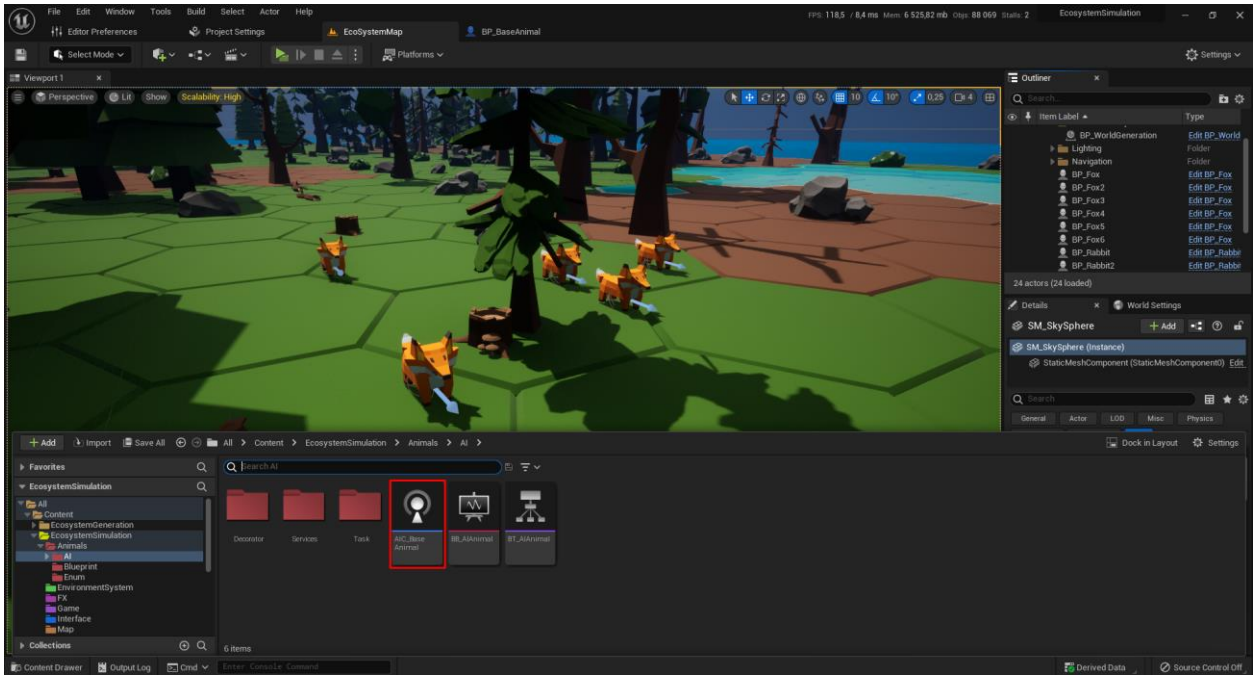
**HealthDecayRate** - *the amount of health that is subtracted with a frequency of "StatsUpdateRate" (starts when "Food" or "Water" is 0);*

**HungerDecayRate** - *the amount of food that is subtracted with the frequency "StatsUpdateRate" (works after the start of the game);*
**ThirstDecayRate** - *the amount of water that is subtracted with the frequency "StatsUpdateRate" (works after the start of the game);*
**StaminaDecayRate** - *the amount of stamina that is subtracted with the frequency "StatsUpdateRate" (works after the start of the game);*
**StaminaRefreshRate** - *the amount of fatigue that is added with the frequency "StatsUpdateRate" (works when the animal is sleeping);*
**MatingRefreshRate** - *the number of units of attraction to the opposite sex that is added with the frequency "StatsUpdateRate" (works at the start of the game, affects the reproduction rate of the species);*

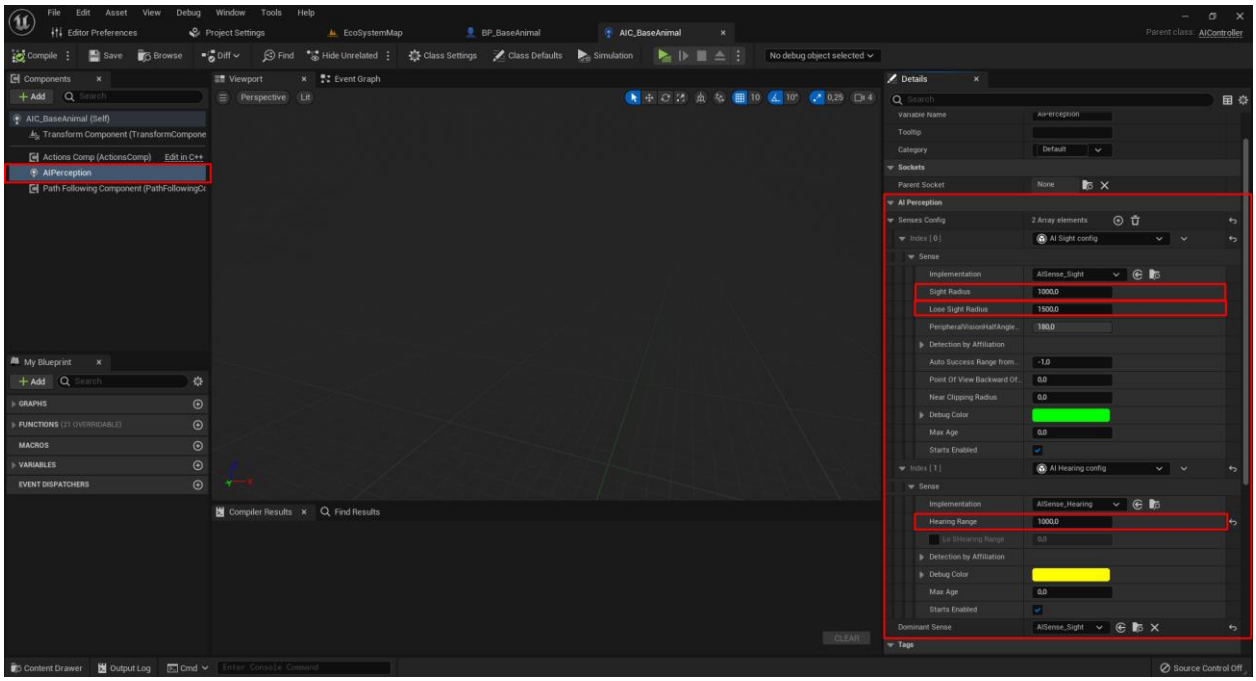
**INFO:**
**IsShowInfo?** - *a parameter that allows you to turn on the information panel above the animal;*
**AnimalName** - *name of a specific animal species;*
**MaxPopulation** - *maximum amount of a particular species;*
**Nutrition** - *parameter of the species belonging to herbivores or carnivores;*

**DETAILS:**
**Age** - *animal age parameter (from 0 to 1);*
**GrowthTime**- *the time after which the animal grows up;*
**BabyActorScale** - *the size of the child character relative to the parent (it is recommended to set from 0.1 to 0.75);*
**IsRandomGender?** - *a parameter that allows you to insert a random gender at the start of the game;*
**Gender** - *the gender of a particular character;*
**BabiesPerLitter** - *a parameter that contains information about the minimum and maximum number of children in a particular species;*

2.3. The next step is to change the settings of the hearing and gaze sensors. These sensors are used by predators to see their prey, and prey to see their predator, and then they flee.



Here, you can experiment with the parameters of hearing and gaze to achieve the desired result.



**Sight Radius** - *the distance at which the animal sees its enemy;*

**Lose Sight Radius** - *is the distance at which an animal loses its enemy;*

**Hearing Range** - *the distance at which the animal can hear its enemy;*

If you have any questions, problems, or need assistance, please send me an email at stylus219@gmail.com.

I hope you have a nice day!